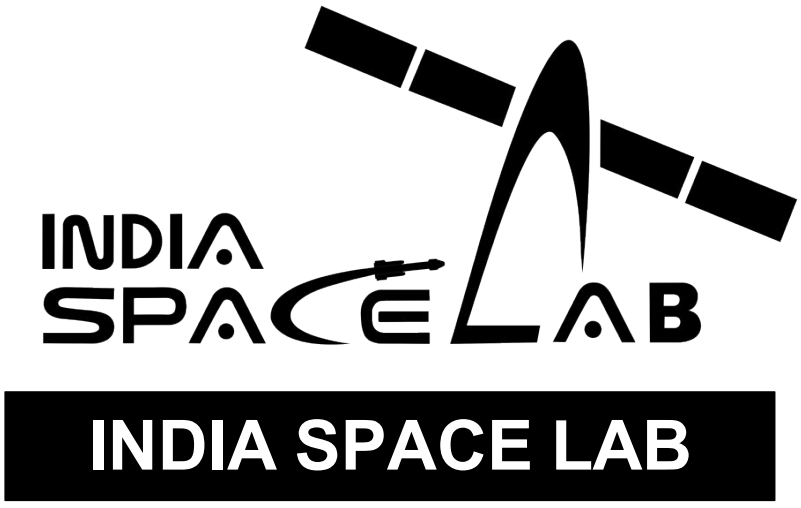
Advanced Drone Technology: Introduction to Guidance and Control Using Python Simulations



INDIA SPACE LAB

Project Report

17th June to 31st July 2026

Student Name: SRINJOY SIL
Institute Name: MCKV INSTITUTE OF ENGINEERING
Institute Roll no.: 11600223119
Enrollment no.: ISL-522235

Advanced Drone Technology: Introduction to Guidance and Control Using Python Simulations

Objective :

The objective of this project is to understand the fundamentals of guidance and control systems through Python-based simulations. The project focuses on tuning a PID controller for drone altitude control and analyzing the trajectory tracking performance of an autonomous boat under both normal and disturbed environmental conditions.

Task 1 - Drone Altitude Control Using PID Controller

Description : The drone altitude control simulation was performed using the provided Google Colab notebook. The Proportional (P), Integral (I), and Derivative (D) controller gains were adjusted to achieve a stable altitude response. Wind disturbance was introduced during the simulation to evaluate the controller's ability to maintain the desired altitude.

Final PID Parameters:

Parameter	Value
P Gain	4.5
I Gain	2.0
D Gain	5.0
Wind Disturbance	5.0

PID Controller Tuning Result Plot:

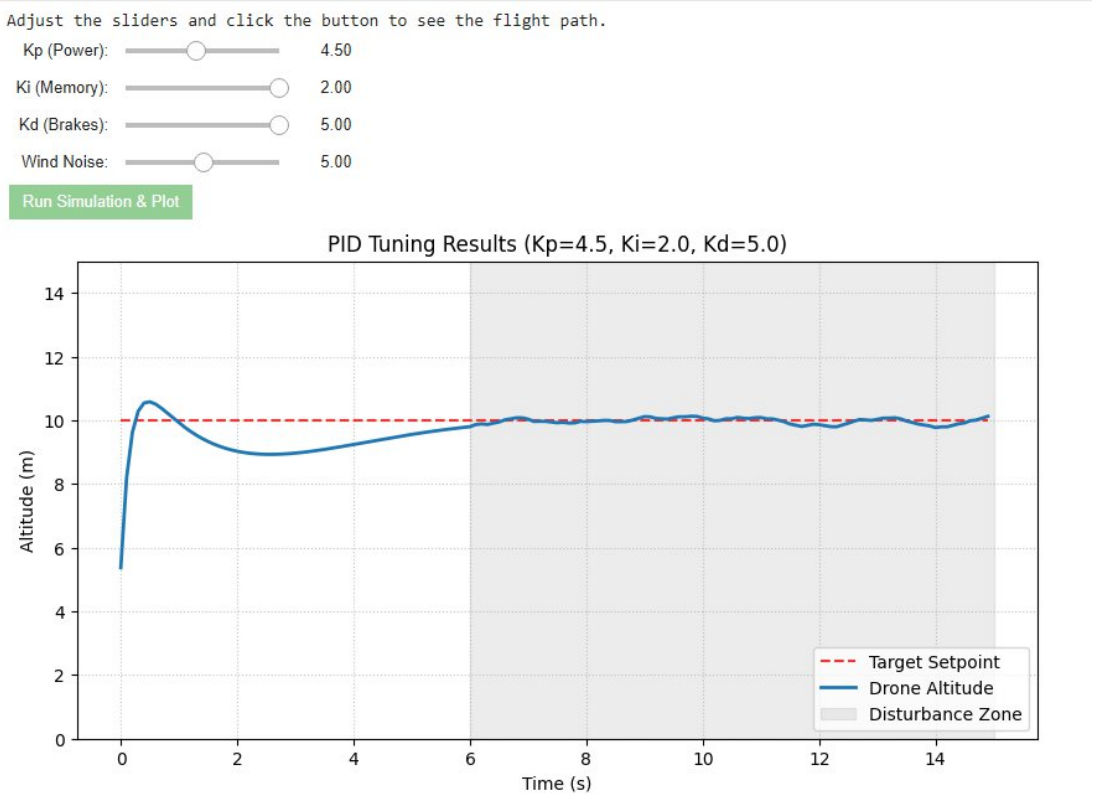


Figure 1: PID Controller Tuning Result Plot

Observation

The tuned PID controller enabled the drone to reach the desired altitude quickly with minimal oscillation. The proportional gain improved the response speed, the derivative gain reduced oscillations, and the integral gain eliminated steady-state error. After the wind disturbance was introduced, the controller successfully stabilized the drone and maintained the target altitude.

Task 2 – Autonomous Boat Guidance

Description : The autonomous boat guidance simulation was executed using the provided Python notebook. The controller gains (kp and kd) were adjusted to improve the trajectory tracking performance of the boat. Simulations were performed under both ideal conditions and in the presence of environmental disturbances to evaluate the robustness of the guidance controller.

Final Controller Parameters:

Parameter	Value
kp	1.0
kd	0.0

Boat Guidance Plot (Without Disturbance)

Environmental Conditions

Parameter	Value
current_x	0.0
current_y	0.0

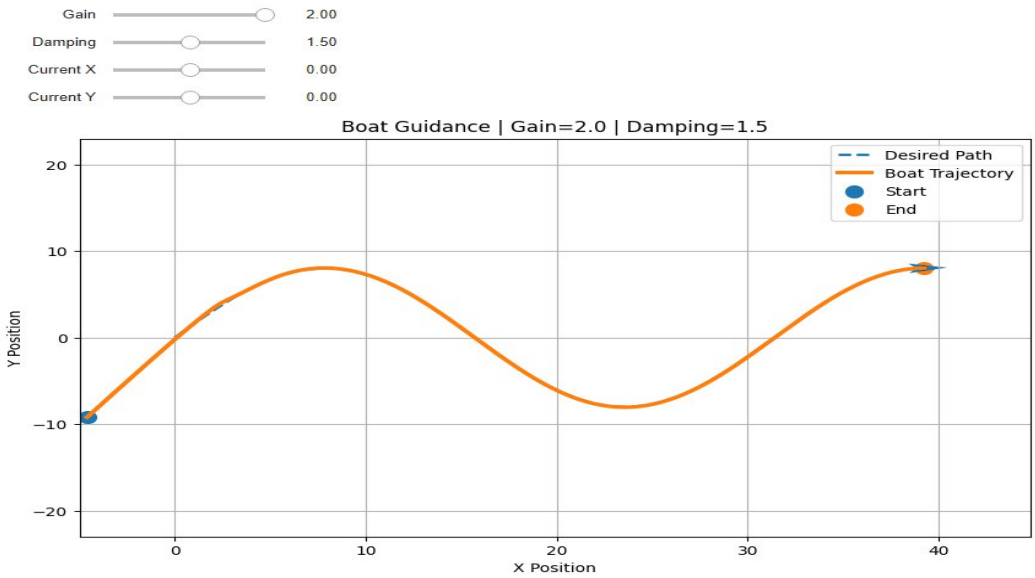


Figure 2: Boat Guidance Plot Without Disturbance

Observation

Without any environmental disturbance, the boat closely followed the predefined reference trajectory. The controller maintained smooth navigation with very small tracking error and demonstrated stable path-following performance.

Boat Guidance Plot (With Disturbance)

Environmental Conditions

Parameter	Value
current_x	0.3
current_y	0.3

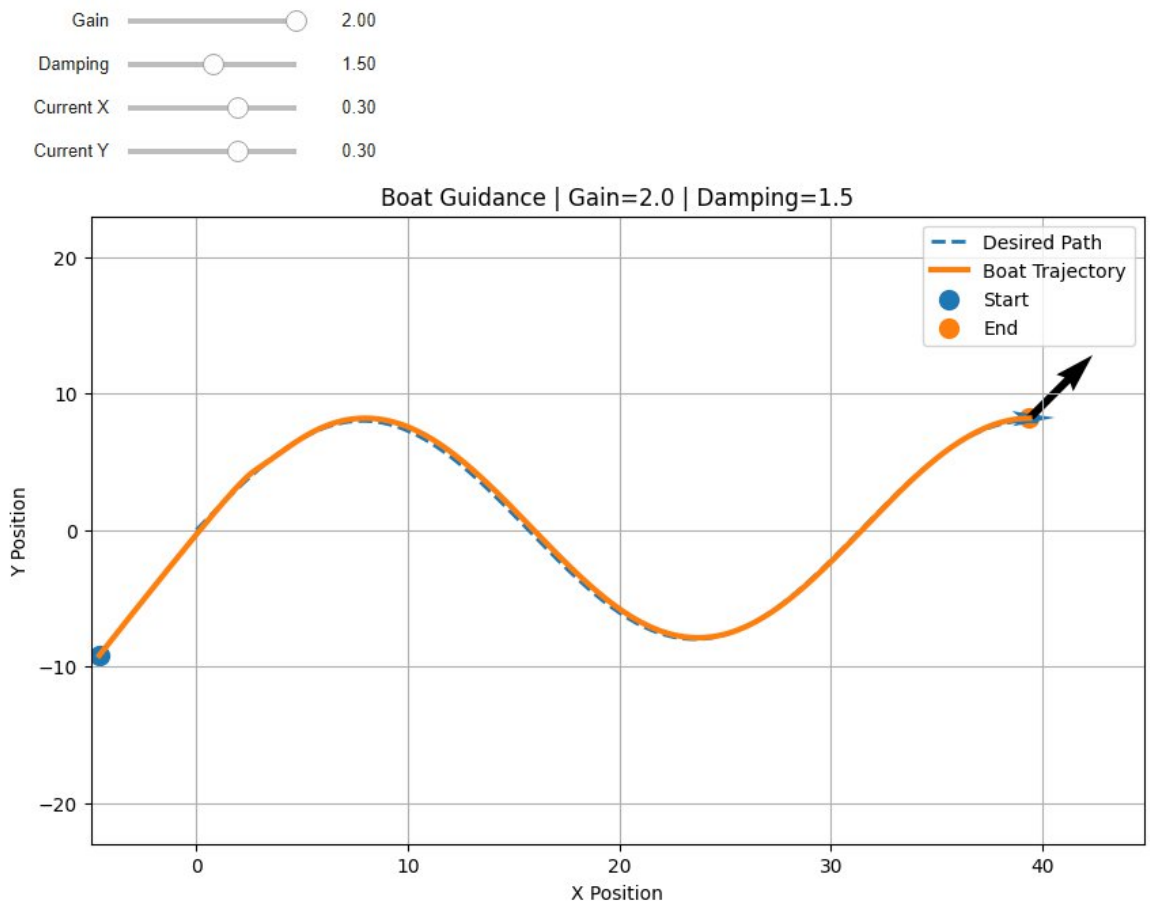


Figure 3: Boat Guidance Plot With Disturbance

Observation

The introduction of water current caused slight deviation from the desired path. However, the controller effectively compensated for the disturbance and gradually guided the boat back toward the reference trajectory, demonstrating good disturbance rejection capability.

Conclusion

This project successfully demonstrated the principles of guidance and control using Python-based simulations. Through PID controller tuning, stable altitude control of the drone was achieved, and the controller effectively handled external wind disturbances. Similarly, the autonomous boat guidance simulation demonstrated accurate trajectory tracking under both normal and disturbed environmental conditions. Overall, the project enhanced practical understanding of PID control, autonomous navigation, feedback systems, and disturbance compensation techniques commonly used in modern unmanned aerial and marine vehicles.